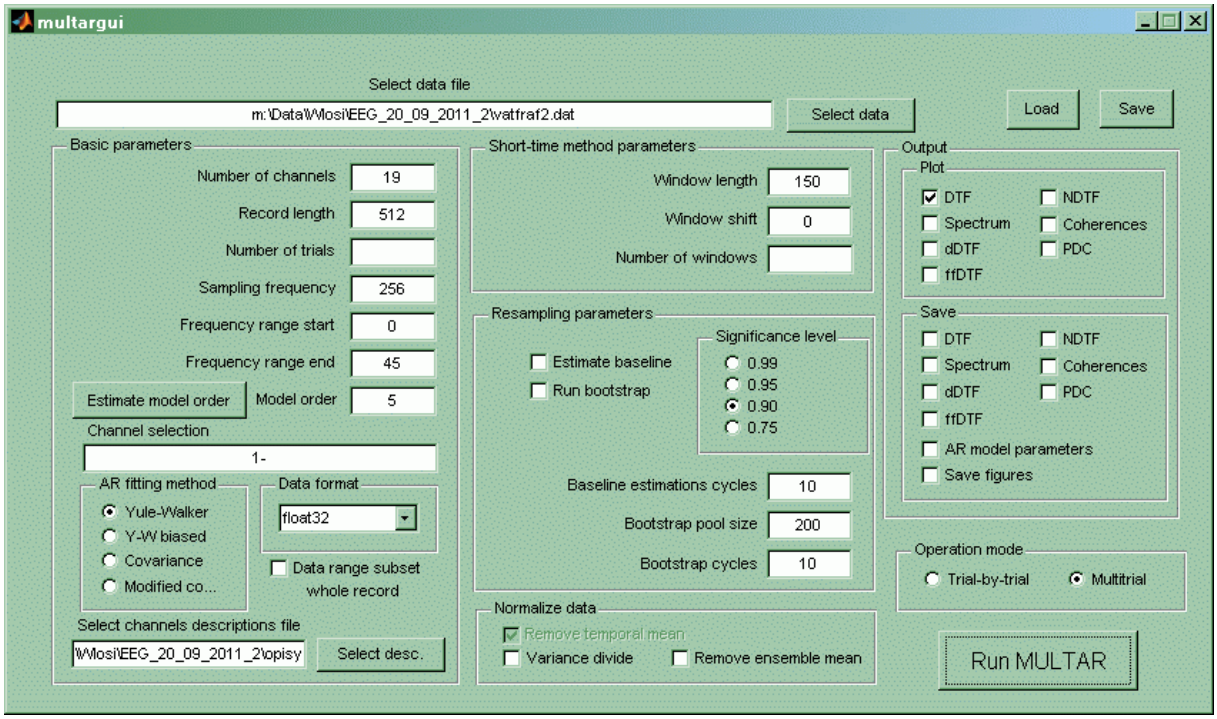
How to use MMULTAR

1. The MMULTAR is an application written in Matlab and it needs the Matlab to be installed on your computer.
2. The MMULTAR package can be downloaded from
   `http://eeg.pl/Software/mmultar_inst.zip`
   For Windows-based systems there is a self-extracting version available as:
   ~~`http://eeg.pl/Software/mmultar_inst.exe`~~
   `http://brain.fuw.edu.pl/~moira/other/MMultar.exe`
3. The downloaded package should be unpacked to an empty directory on the destination computer. For the purposes of this manual we will assume that the package was placed in `C:\Program Files\Mmultar` directory.
4. Start Matlab, select from menu bar File→Set Path and press the “Add Folder...” button. Select your MMULTAR directory and press “Save”.
5. Start the package by typing in the Matlab command window:
   `mmultar`
6. The main dialog window should appear:



Program parameters

Datafile

- Select data file: this editbox allows to enter the name of the data file to be processed. You may enter the name from the keyboard or use the “Select data” button in the upper right part of the window. A standard open file dialog window then appears allowing to select a file on disk. After successful selection the file name is visible in the edit box left on the Select data button.
- Data format: Basically, the program allows to read raw binary data. The values can be encoded as IEEE single or double precision floating points (float32, float64), 16-bit

signed integers (int16) or 32-bit signed integers (int32). You can also import an ASCII data file with each channel values in a separate column.

- Number of channels, record length and number of trials (repetitions of each record): these parameters must be entered manually for raw binary files with no header. One of these values may be omitted and it will be calculated from the remaining two, the data format and the datafile size.
- If the “Data range subset” checkbox is checked, you can enter a time samples subset. The original data records will be truncated to the entered range (instead of using the whole records stored in file). All the calculations will be conducted on truncated data.

Additional info

- Sampling frequency of the original data.
- Channel selection: we may use only a subset of channels from all channels stored in the original file. By entering the desired selection in the “Channel selection” edit box we are able to change the set of active channels or change their order. The default selection is “1–“ which means “from channel 1 to the last one”.
  Example: “3-5,2,6,7-“ means channels 3, 4, 5, 2, 6 and from 7 to the last one.
- Select channels descriptions file: if we want to get channels labeled by their names on the results printouts, we should provide the name of text file containing channel labels (one in each line) by pressing button “Select desc” near “Select channel description file” editbox and choosing the file on disk.
- Input data normalization: program assumes the data in each channel has zero mean. This option is always used and the checkbox has only informative role. You may experiment with dividing each channel by its variance. For multitrial data it is possible to remove the ensemble mean (“ERP” calculated for each datapoint in every channel).

AR model info

- The DTF method is based on the autoregressive model fitted to the data. The optimal model order may be chosen manually, in the “Model order” edit box. Alternatively we may click the “Estimate model order” togglebutton. In its pressed state the model order criteria will be calculated first, helping in choosing the proper model order.
- The specific AR fitting algorithm may be selected in the section “AR fitting method” by clicking the appropriate radiobutton. Note: the “Yule-Walker biased” option is currently inactive.
- “Frequency range start” and “Frequency range end”: The results will be presented in the frequency domain. We should provide the data sampling frequency and minimal and maximal frequency of the calculated functions (“Frequency range start”, “Frequency range end”). All results will be functions over frequency within the specified range.

Short-time method parameters

- Window length: the size (in samples) of data window
- Window shift: the number of samples to shift the next data window
- Number of windows: number of windows to calculate.

Operation mode

- Trial-by-trial: in this mode program calculates results for each trial separately.
- Multitrial: in this mode all trials are used to produce one AR model. Normally, the whole record will be used as one window. If number of windows is put as 1 (in "Short-time method parameters" section), the "Window length" (if present) will be used and the data record length will be adjusted accordingly.

Resampling statistics

- Estimate baseline: the data is randomly shuffled in each channel and the functions are calculated. The shuffling and estimation is repeated "Baseline estimation cycles" times. This allows to estimate the baseline (zero) level for every function. For the 0.90 significance level the 0.90 percentile is used. Note: one baseline level is calculated for the whole frequency range.
- Run bootstrap: for the mutlitrial data new sets of trials are randomly drawn from the original set of trials. For each new set of trials all functions are calculated. After "Bootstrap cycles" repetitions of that process the appropriate percentiles are calculated from the so created distributions (e.g. for the 0.90 significance levels percentiles 0.05 and 0.95 are used).
- Bootstrap pool size: the size of trial sets used in bootstrap calculations. Basically it should be equal the number of trials.
- Significance levels: the statistical significance limits for baseline and bootstrap results.

Output

- Plot: a set of checkboxes indicating which functions will be calculated and plotted
  - DTF: normalized directed transfer function;
  - NDTF: non-normalized directed transfer function;
  - Spectrum: power spectral density (modulus);
  - Coherences: modulus of: ordinary (above diagonal), multiple (on the diagonal) and partial (below diagonal) coherences;
  - dDTF: direct directed transfer function;
  - ffDTF: full-frequency directed transfer function;
  - PDC: partial directed coherence;
- Save: a set of checkboxes indicating which functions will be calculated and saved on disk. The set of functions similar to the "Plot" section above. The functions are saved as Matlab files allowing further processing of the result. At this moment there is no option to reread them in Mmultar and show on the screen. "AR model parameters": the estimated model parameters and transfer matrix (**H**) will be written to disk. "Save figures": when checked every new figure is automatically saved on disk (as TIF image).

Load and save

- "Load" and "Save" buttons are used to store ("Save") and retrieve ("Load") the complete set of parameters for the whole Mmultar window. Filenames, checkboxes and radiobuttons status and entered values can be saved on disk for later use. Saved sets are stored in Matlab format and can be retrieved using "Load". Both buttons open standard dialog window for disk file name.

Program execution

After entering program parameters the calculations start after pressing the “Run Multar” button in the bottom right corner of the window.

If the “Estimate model order” button is in its pressed state, first the graph of model estimating criteria will be shown. We can select between AIC and FPE criterion, looking for the first minimum of the functions. The model order selected in the upper right edit box will be used in calculations.

Every function is plotted in separate window as a matrix of graphs. Each graph represents a connection between (or “from” in a case of directed quantities) channels below the given column and (“to”) the channel marked left of the given row. The values are plotted over frequency in the range selected by the user (“Frequency range start” and “Frequency range end”). The vertical scaling of the graphs in each window can be changed by pressing the button “Redraw”. The maximal value is written at the bottom of each window to help estimating the proper scaling. The “Set default” button resets the vertical scaling to its default state. Moreover, figures are equipped with “Save” buttons which allow to save them on disk when needed

If you select “Trial-by-trial” mode of operation and the data file contain more than one trial, after plotting the last function a dialog window will be shown asking to proceed to the next trial or quit calculations (by pressing the appropriate button). In that case the program uses the same model order as chosen for the first trial.

In “Mutitrial” mode the output is similar to “Trial-by-trial” mode, but the program produces only one set of results.

When “Window shift” is bigger than 0 and “Number of windows” is bigger than 1 the program starts calculations in short-time mode. Results are displayed as matrices of time-frequency color maps (time on horizontal axis, frequency on vertical axis), organized similarly to the previously described graphs. If “Calculate baseline” is on, the graphs are plotted only for values above the calculated threshold; the values below the threshold are left white. If “Run bootstrap” is on, you can select your picture from three possible sets calculated for the lower ad upper percentile (according to the selected significance level) and for the median (50% percentile), by clicking the appropriate radiobutton below the graph.